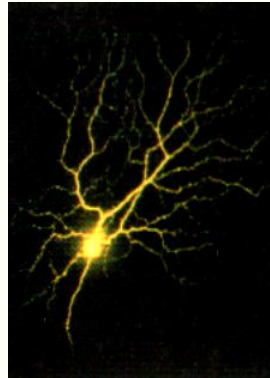


Basic Principles of Medicine II

Module: Nervous System & Behavioral Science

Lecture No; NB 05

Neurons & Glia



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Recommended Reading

'Essential Neuroscience' - by Allan Siegel and Hreday N. Sapru. Wolters Kluwer Health, Inc., 4th edition, 2019.
Chapter 4.

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=196986417&bookid=2482>

Lecture Objectives

- SOM.MK.I.BPM2.1.NB.1.NSCI.0010 Identify two major cell types in the nervous system
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Major nervous system cell types: Neurons and Glia

- Neurons
 - **Excitable**: Generate and conduct action potentials (APs)
 - Main role is **signaling** – integration, communication
- Glia
 - **Supportive** of neurons but also with signalling roles
 - Not excitable in the classic sense (i.e., **do not fire APs**)
 - But oscillations in intracellular Ca^{2+} (calcium waves) promote release of gliotransmitters (e.g., glutamate, ATP)
- Cell types are categorized according to morphology, location and functional properties

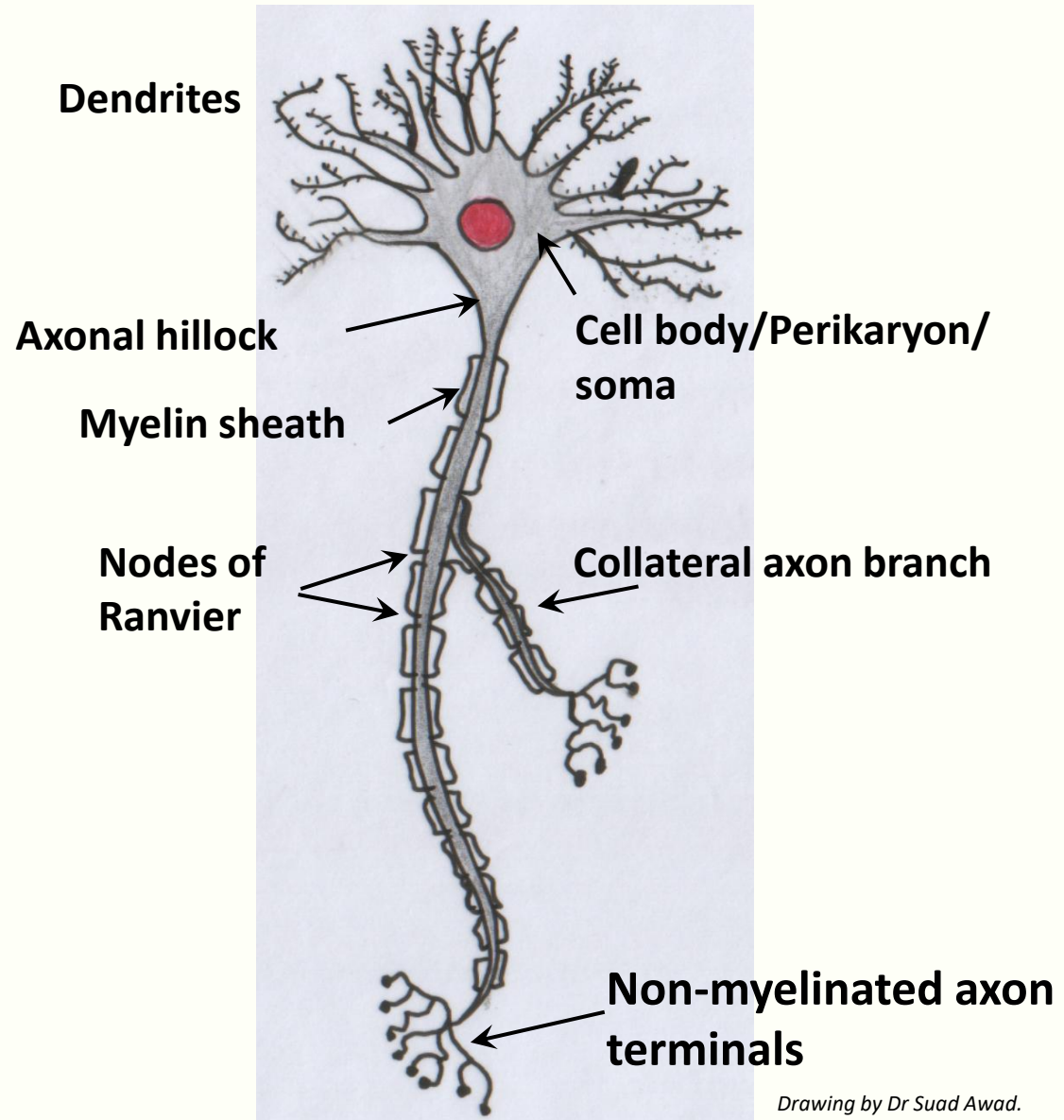
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Basic Structure of Neurons

Morphology is related to function

- Dendrites covered in dendritic spines make synaptic connections with other neurons
- Nissl bodies in the soma contain RNA granules and ribosomes
- Axons and dendrites are collectively known as **neurites**



Drawing by Dr Suad Awad.

Morphology is related to signal reception and transmission: Taking a multipolar neuron, such as a motor neuron as our example, we see the soma expressing with dendrites that may be covered in spines. The primary function of dendrites is to integrate chemical signals emitted from axons (typically from other neurons). The spines, if present, can serve as post-synaptic targets.

Nissl bodies are the histological sign of the rough endoplasmic reticulum, a major site of protein synthesis.

The axon hillock (AH) is the site of initiation of action potentials. In many neurons, the AH arises from a conical elevation of the soma. Unlike other eukaryotic cells, neurites do not readily support diffusion, so transport mechanisms are required for the sufficient movement of materials throughout the lengths of the axons and dendrites.

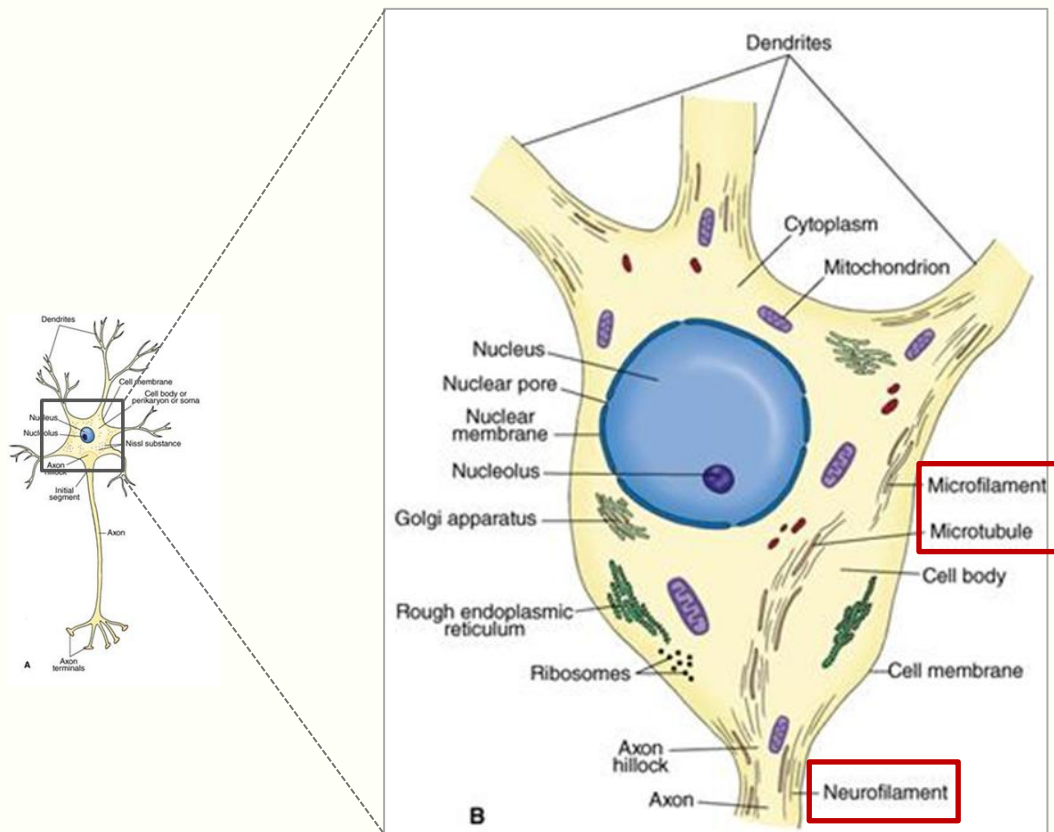
Myelin is fused membranes of Schwann cells or oligodendrocytes forming an insulating sheath around axons, which may collateralize extensively.

Synaptic boutons contain synaptic vesicles with stored neurotransmitter destined to undergo exocytosis. In neurons that release catecholamines, serotonin or peptides there are dense cored vesicles whereas in those like motor neurons that release acetylcholine, the vesicles are clear.

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Components of the Cytoskeleton



- **Microtubules:** dimers of α and β tubulin added at the positive end of microtubules = Polymerization (involved in transport)
 - **MAPs** (microtubule-associated proteins) stabilize microtubules
- **Neurofilaments (NFs)** determine axonal diameter
- **Microfilaments (MFs)** mediate growth cone advance during growth or repair after injury

Cytoskeletal components

- Crucial for neuronal growth and shape and for transport between soma and neurites
- Microtubules
 - Formed from paired of α - and β -tubulin
 - Polar structure creates a positive and minus end of the polymer
 - Stabilized by microtubule-associated proteins (MAPs), e.g. tau proteins
- Neurofilaments
 - Most common filaments in neuron
 - Very stable with little turnover
 - Create the scaffolding of the cytoskeleton
 - Level of neurofilament gene expression controls axonal diameter
- Microfilaments
 - Braids of two thin strands of actin filaments
 - Anchored to membrane by mesh just beneath the cell membrane
 - Participate in growth cone advance during neuronal growth or repair

Lecture Objectives

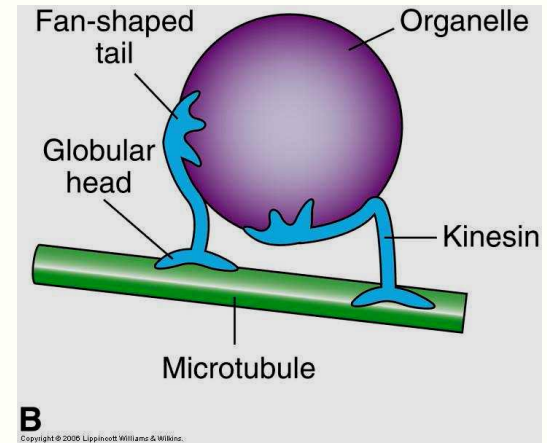
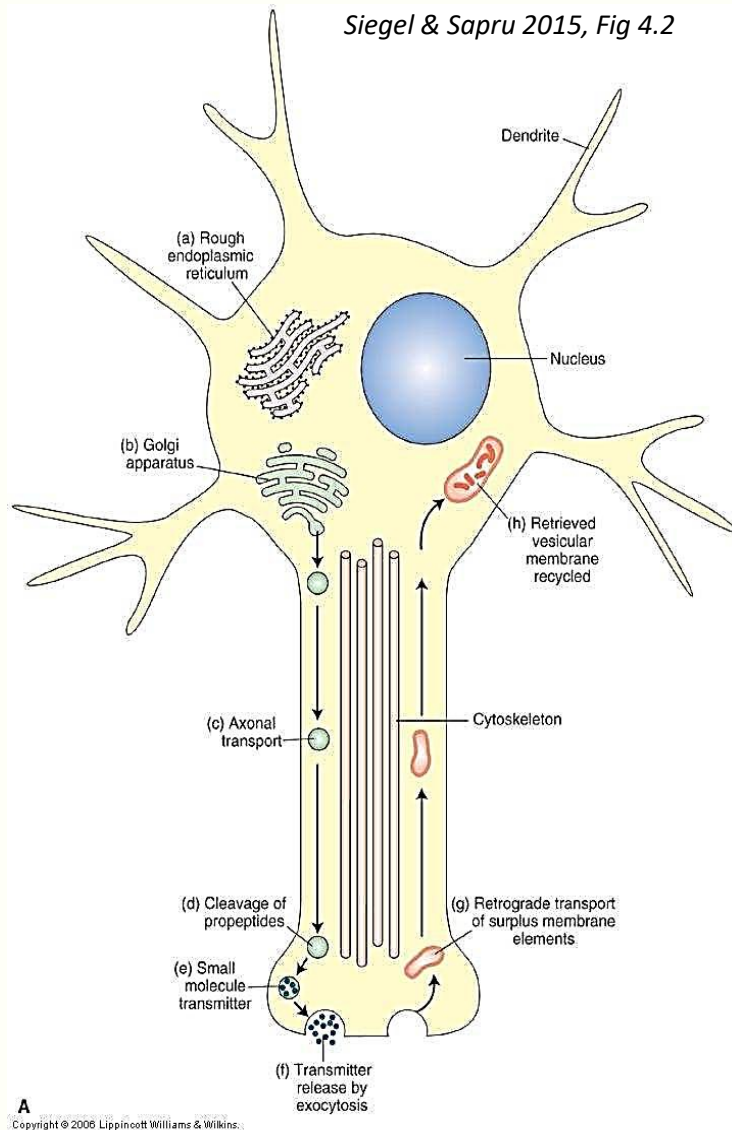
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Axonal Transport

Anterograde transport

Kinesin

(microtubule-associated ATPase)
moves vesicles along microtubules towards
positive end - *away from soma*



Retrograde transport

Dynein

protein (MAP)-ATPase - towards
negative end of microtubules -
towards soma

Transport proteins

- Kinesin
 - ATPase responsible for fast anterograde transport
 - Similar to myosin in muscle
 - Binding sites for attachment of large structures, e.g., vesicles and mitochondria
 - Movement towards MT positive end, i.e., nerve ending
 - Becomes inactivated in nerve ending and carried back to the soma by retrograde axonal transport to be re-activated
- Dynein
 - ATPase responsible for fast retrograde transport
 - Movement towards MT negative end, i.e., soma
 - Inactivated in soma, activated again in nerve ending

Axonal Transport

Type of transport	Speed (mm/day)	Material transported
Anterograde (Kinesin-driven)	Fast (100-500)	Vesicles, Mitochondria
	Slow (1-10)	Soluble proteins (calmodulin) and enzymes
	Slow (0.1-1.0)	Cytoskeletal molecules (Tubulin, Actin, NF protein)
Retrograde (Dynein-driven)	Fast (200-300)	Lysosomes, enzymes, recycled vesicular membranes, NGF Viruses: Herpes, Rabies

Dendritic transport, axonal flow

- **Transport along dendrites**
 - Axonic microtubules always have negative end towards soma and positive end distally
 - Dendritic microtubules are of mixed orientation
 - Half with positive ends facing soma, other half opposite
 - This may selectively direct movement of some materials to dendrites rather than down axons

Axonal Transport Occurs at Different Rates

Without fast axonal transport, the soma and nerve ending would be unable to exchange materials at a satisfactory rate to retain the viability of the distant endings of the neuron. By contrast, the neuron's electrical signalling ability is **fast** over the same distance.

Retrograde transport for lysosomes and endolysosomes degrades endocytosed material and redundant cellular components are degraded. These materials are packaged in large membrane-bound organelles that are part of the lysosomal system.

Retrograde Axonal Transport: Some viruses use retrograde transport to infect neurons. Examples are the herpes virus and the rabies virus which gain access to neuronal cell bodies by entry into axons in the skin. The rabies virus replicates well in the cell body, destroying its host cell and then passing to nearby neurons to continue its devastating work.

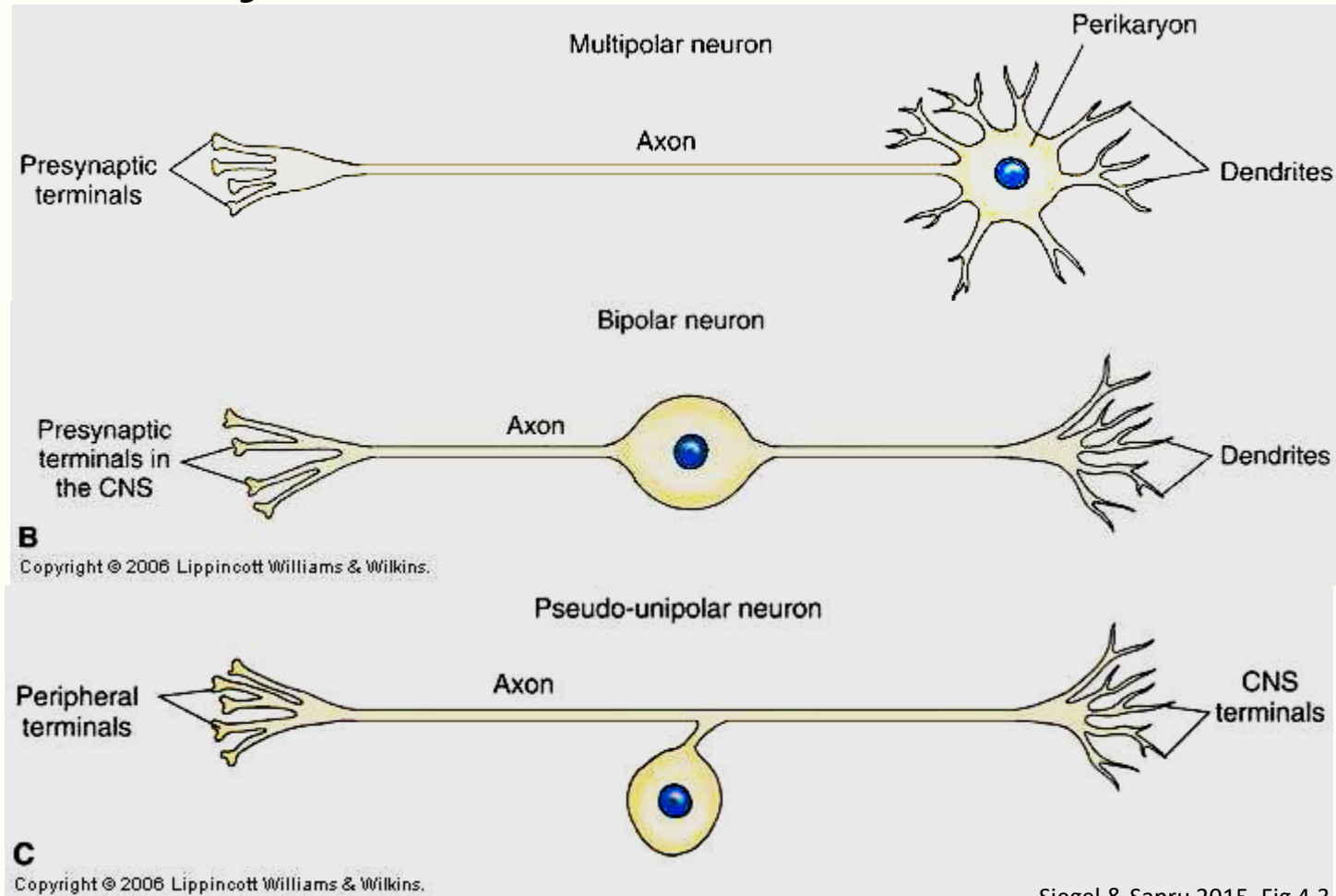
Interruption of axonal transport leads to death of axons distal to the site of injury in a process called Wallerian degeneration occurs, likely in relation to disrupted axonal transport mechanisms. In Alzheimer's disease, the microtubule-associated protein Tau is deranged, leading to the intracellular formation of neurofibrillary tangles, thereby disrupting microtubular structure and axonal transport.

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Morphotype Influences Signal Reception and Transmission

Number of Neurites



Siegel & Sapru 2015, Fig 4.3

One way to classify a neuron is by the total number of neurites emerging from the soma. One class of neurite is the **dendrite** and the other is the **axon**. The morphological types are pseudo-unipolar, bipolar and multipolar, the names relating to the pattern of neurites extending from their cell bodies.

Single neurite = pseudounipolar, two = bipolar, three or more = **multipolar**.

Most neurons in the brain are multipolar. **Motor neurons** in the spinal ventral and lateral horns of the SC, autonomic ganglion cells. Mostly inhibitory in the CNS and referred to as Golgi type II. Multipolar neurons with long axons are called Golgi Type I; multipolar neurons with short axons are often inhibitory and are called Golgi Type II.

Bipolar - retinal, olfactory and acoustic. Have two axonal processes that emerge from the cell body.

Pseudo-unipolar spinal sensory neurons have one process, which divides into a central branch and a peripheral branch. Each branch has structural and functional characteristics of an axon. They are called pseudo-unipolar because they are originally bipolar. The two processes fuse during development to form a single process that bifurcates at a distance from cell body. These are peripheral sensory neurons, the cell bodies of which aggregate to form **spinal** and **cranial nerve ganglia**.

Functional Classification of Neurons

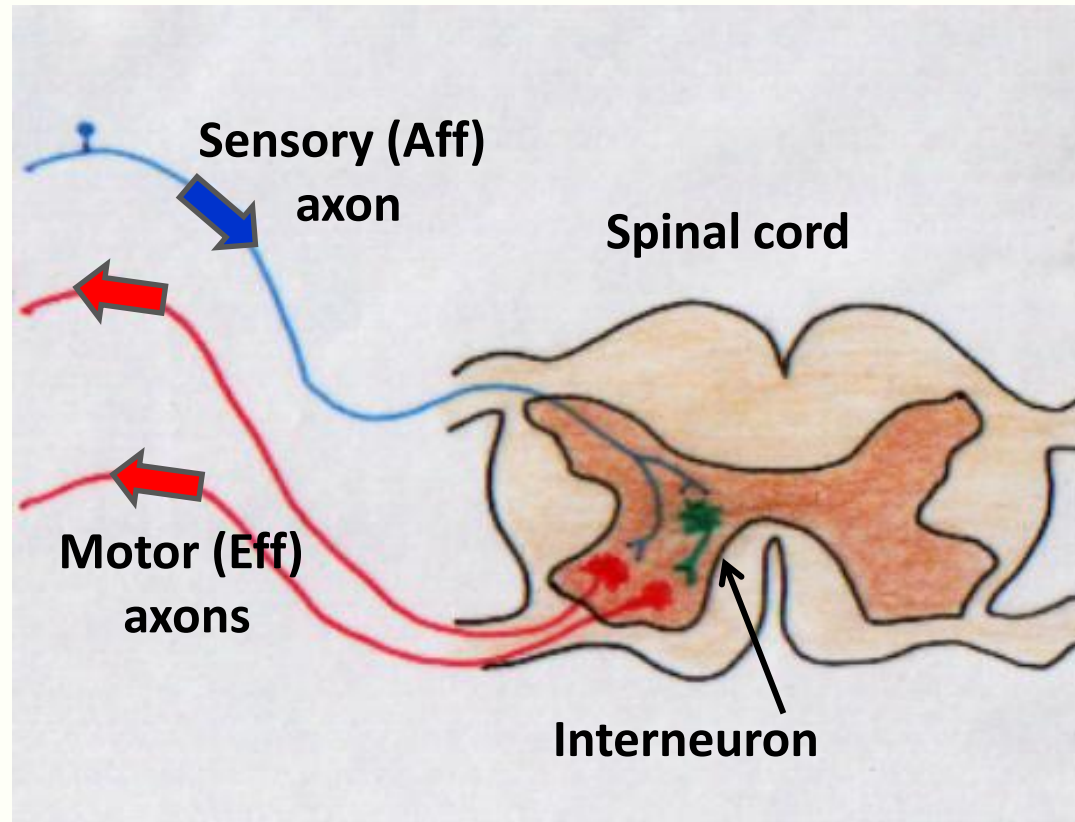
Afferent (eg sensory)

From PNS to CNS



Efferent (eg motor)

From CNS to PNS



Interneurons



Connect and modulate input-output of
Afferent and *Efferent* neurons in CNS and
PNS

- Afferent is a generic term, referring to the movement of information **TO** some particular destination
 - But, the term is commonly synonymous with primary sensory neurons/fibers
- Efferent is also a generic term, referring to the movement of information **AWAY** from some particular source
 - But, the term is commonly synonymous with peripheral motor neurons/fibers
- Interneurons are neurons that serve as intermediaries between other neurons
 - The vast majority of neurons are interneurons

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Characterization of axons by myelination and diameter

- Axons are commonly myelinated (electrically insulated by specialized glial cells)
 - Myelination speeds axonal conduction of action potentials and reduces energy needs for ionic concentration gradients
- Neurons can be grouped or categorized according to their axonal diameter and myelination
 - Both correlate with conduction velocity
 - Class I: Largest and most rapidly conducting fibres
 - Classes II and III: Intermediate diameter
 - Class IV: Unmyelinated and smallest diameter

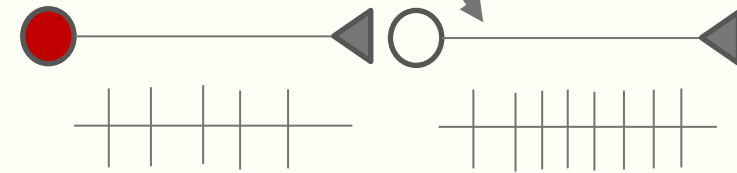
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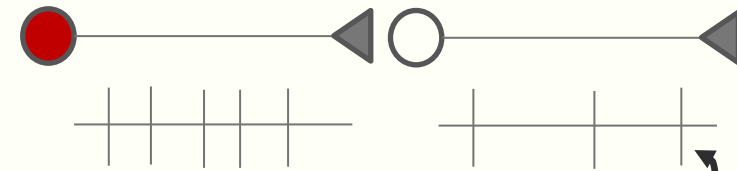
Excitatory and Inhibitory Neurons

- Excitatory neurons
 - Cause post-synaptic excitation (depolarization)
- Inhibitory neurons
 - Cause post-synaptic inhibition (hyperpolarization)
- Transmitter effect **depends on post-synaptic receptors**
 - e.g., dopamine D₁ family coupled to G_{sα} → ↑[cAMP]
 - vs. dopamine D₂ family coupled to G_{iα} → ↓[cAMP]

Increased frequency of action potentials



**Postsynaptic excitation =
*Depolarization***



**Postsynaptic inhibition =
*Hyperpolarization***

Decreased frequency
of action potentials

Not uncommonly, neurons are described as either excitatory or inhibitory. Transmitter released by excitatory neurons tends to depolarize membranes of other neurons; transmitter released by inhibitory neurons tends to polarize or stabilize membranes of other neurons. It is noteworthy, however, that receptors (not transmitters) ultimately dictate excitatory versus inhibitory effects of transmitters (or the neurons from which the transmitters are released).

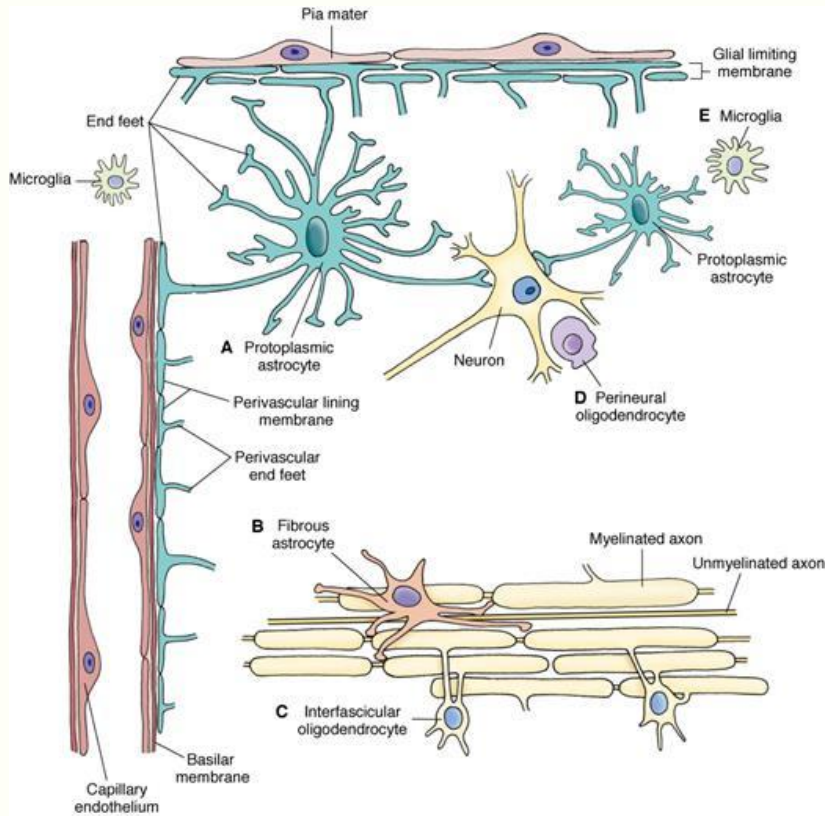
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Neuroglia

- *Glia = Greek for glue*
- Glial cells outnumber neurons in the nervous system
- Key roles:
 - Maintaining the ionic milieu of nerve cells
 - Modulating the rate of nerve signal propagation (myelination)
 - Modulating synaptic activity by controlling the uptake of the neurotransmitters at or near the synaptic cleft
 - Providing a scaffold for neural development (radial glia)
 - Participating in recovery from neural injury
- Four main types in CNS:
 - Astrocytes
 - Oligodendrocytes
 - Microglia
 - Ependymal cells

Astrocytes



Key characteristics:

- Restricted to brain and spinal cord (CNS)
- Most common CNS glial cell type

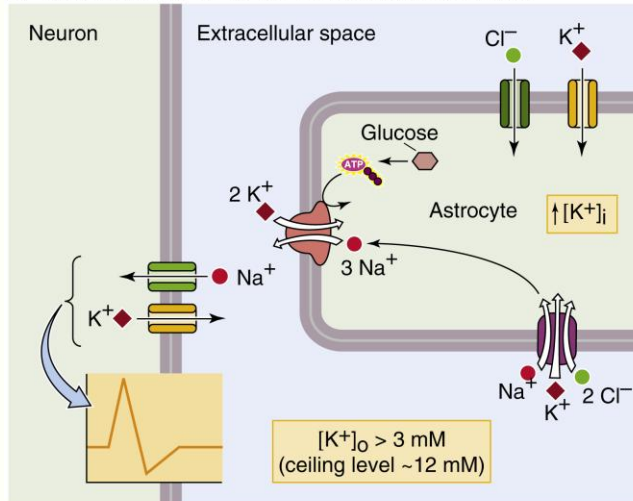
Key functions:

- Component of blood-brain barrier
- Scar formation
- Regulate chemical content of extracellular space
 - Limit spread of neurotransmitter from synapses
 - Uptake of neurotransmitters after release from nerve terminals
 - Recycling neurotransmitters such as Glu, GABA
- Uptake of excess K^+ released following APs
 - Disperse K^+ throughout a greater volume: spatial buffering

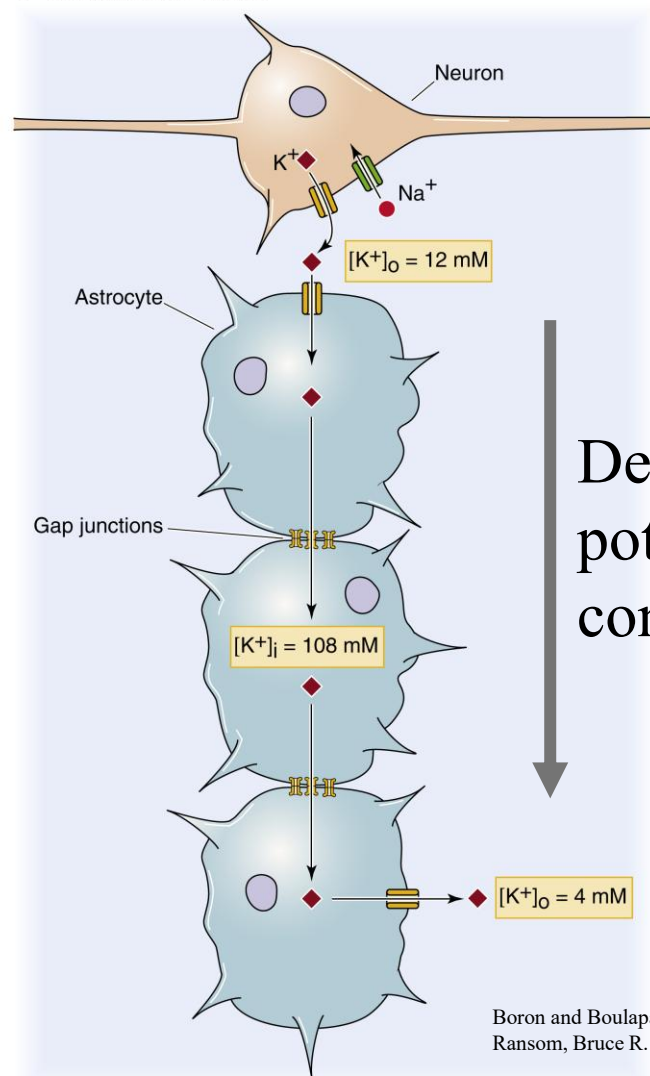
- **Astrocytes:**
 - Restricted to CNS
 - Elaborate local processes that give the cells star-like appearance (hence the prefix astro)
 - Functions
 - Maintain chemical environment for neural
 - Restrict spread and lifespan of transmitters
 - Uptake of excess extracellular K^+ released following nerve impulses in the EC space

K⁺ handling by astrocytes

A—MECHANISMS OF K⁺ UPTAKE BY ASTROCYTES



B—SPATIAL BUFFERING



Astrocytes and K⁺ regulation

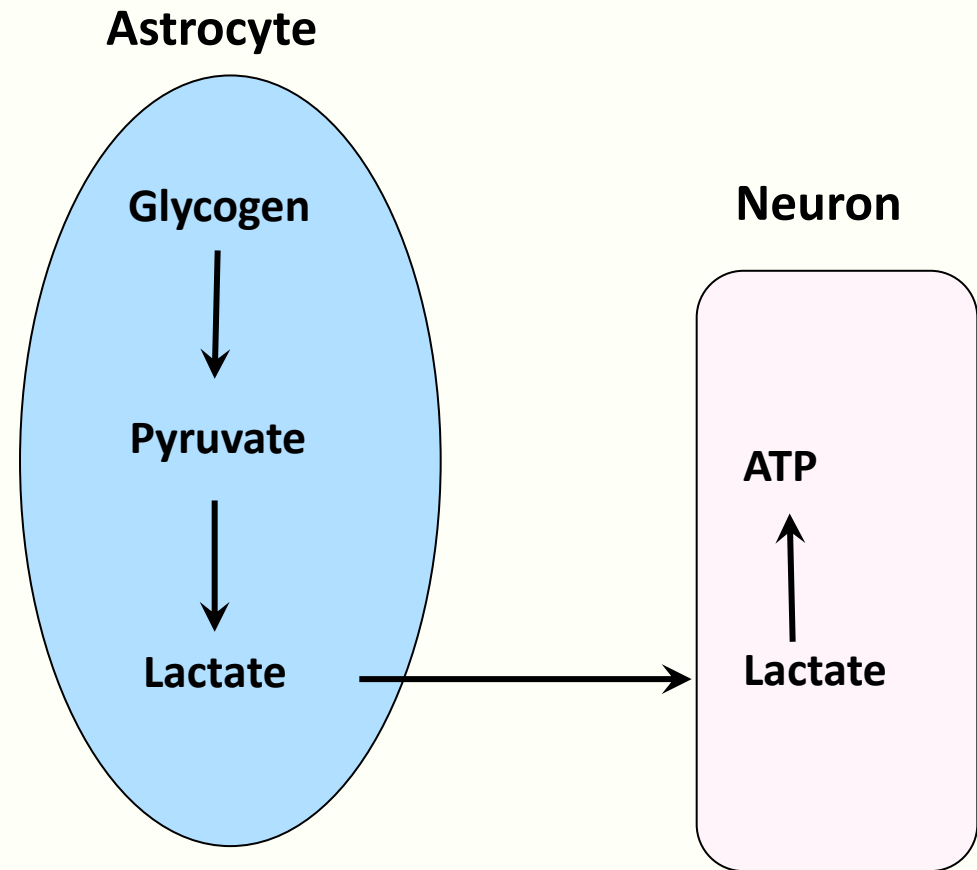
- [K⁺]_o increases after impulse propagation in neurons
- High [K⁺]_o interferes with AP generation
- Excess K⁺ is taken into astrocytes via
 - Na/K ATPase
 - Na/K/Cl⁻ co-transporter (NKCC)

Spatial Buffering

- Astrocytes then distribute K⁺ throughout a greater volume
- Range enhanced by gap junctions between astrocytes

Neuronal energy support

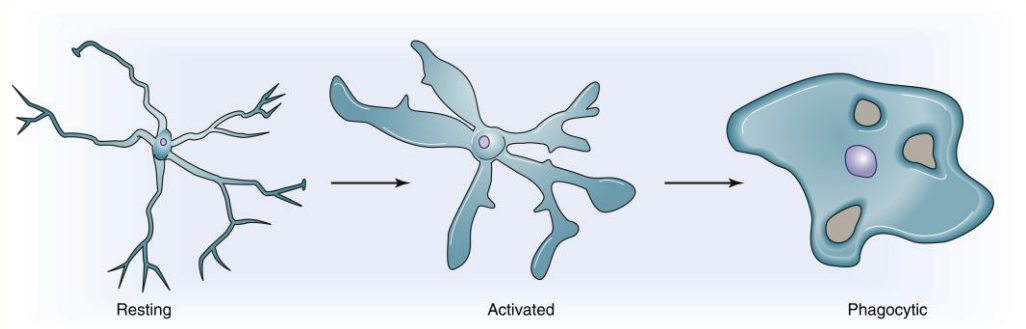
- Main source of neuronal energy is circulating glucose
- Astrocytes store glycogen and supply neurons with lactate
 - Alternative energy source for ATP generation
 - Exhausted within ~10 min



- **Astrocytes store Glycogen and Supply Neurons with Lactate**
 - In the CNS, astrocytes are the main storage depot for glycogen which can be converted to lactate and transferred to neurons as an alternative source for ATP production. Neurons pick up extracellular lactate and use it to generate energy in the form of ATP.
 - If blood supply is interrupted, the normal supply of glucose to the CNS will be limited. Neuronal activity can survive this loss but only until the glycogen is used up (within about 10 minutes of blood supply interruption).

Microglia

- CNS macrophages
- Smaller than astrocytes/neurons
- Key functions:
 - Phagocytosis
 - Removal of debris
 - Synaptic pruning
 - Immune activation
 - Produce pro-inflammatory cytokines during neuronal injury
- Two main states
 - Resting/surveillance: extensive branching processes
 - Activated: rounded, possibly with phagocytic inclusions

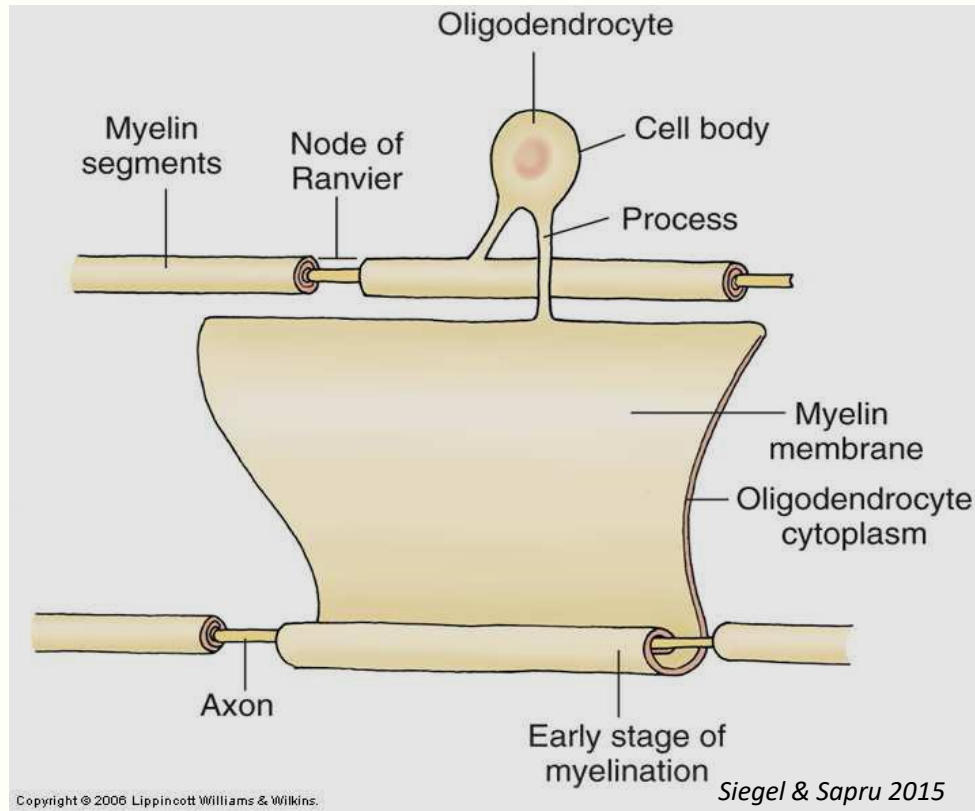


Boron and Boulapaep Concise Medical Physiology
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Microglial cells: Derived primarily from hematopoietic precursor cells rather than ectodermal cells (although some may be derived directly from neural precursors). They share many properties with macrophages found in other tissues and may be classified as such. They are primarily scavenger cells that remove the cellular debris from the site of injury. Additionally, they secrete signaling molecules (cytokines) that can modulate local inflammation and influence the survival or death of cells. Following brain injury, the number of microglia increases dramatically. Some proliferate from microglia resident in the brain, while others come from cells that migrate to the site of injury and enter the brain via local disruption of cerebral vasculature.

Microglia are constantly moving and analyzing the CNS for damaged neurons, plaques, and infectious agents. The brain and spine are considered “immune privileged” organs in that they are separated from the rest of the body by a series of endothelial cells and astrocytes, which prevents most infections from reaching the otherwise vulnerable nervous tissue. In the case where infectious agents are directly introduced to the brain or cross the blood-brain barrier, microglia must react quickly to increase inflammation and destroy the infectious agents before they damage the sensitive neural tissue. Due to the unavailability of antibodies from the rest of the body (antibodies are too large to cross the BBB), microglia must be able to recognize foreign bodies, swallow them, and act as antigen-presenting cells activating T-cells. Since this process must be done quickly to prevent potentially fatal damage, microglia are extremely sensitive to even small pathological changes in the CNS. They achieve this sensitivity in part by having unique potassium channels that respond to even small changes in extracellular potassium. When the CNS is injured, microglia become enlarged, mobile and phagocytic.

Myelinating Glia



- **Oligodendrocytes** are found in the CNS
- **Schwann cells** are the myelinating glia of the PNS
- Myelination provided by flattened cytoplasm, wraps up to 100 membrane/myelin layers around axons
- Interrupted by nodes of Ranvier ($1\mu\text{m}$) every $1000\mu\text{m}$

Oligodendrocytes: In the CNS, oligodendrocytes wrap their membranes around long segments of axons to lay down myelin composed of up to several dozen membrane layers. The myelin acts as an electrical insulator and reduces capacitance of the axonal membrane and thereby elevates conduction velocities of impulses. The small regions (1 micrometer) of unmyelinated membrane between adjacent myelinated segments are called the Nodes of Ranvier. Oligodendrocytes myelinate several segments on the same axon and segments of several axons in its vicinity. Each myelinated segment is about 1000 μm long.

Schwann Cells: In the PNS, Schwann cells myelinate axons. Each Schwann cell myelinates only one segment of an axon. So, each myelinated peripheral axon is insulated by a large population of Schwann cells. During myelination in the PNS, Schwann cells rotate around the axon.

Schwann Cells Facilitate Regeneration of Severed Axons

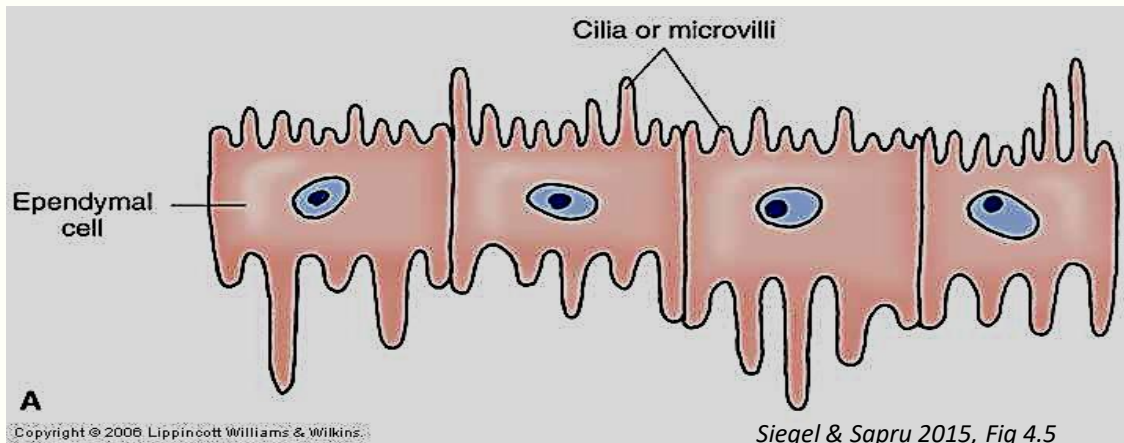
When an axon is cut in a peripheral nerve, Schwann cells retract from the isolated distal axonal segment, increase in number and form a guide tube. They also secrete growth factor to induce axonal growth of the proximal axonal segment along the guide tube to its target cell.

The Axon Instructs the Schwann Cell to Myelinate It

The axon destined to become myelinated sends an appropriate signal to the Schwann cell to induce wrapping and myelination. Small diameter axons introduced experimentally to Schwann cells do not induce them to form myelin.

Ependymal cells

- Originate from ventricular zone in the neural tube
- Two types
 - **Choroidal Ep cells** in choroid plexus: CSF secretion
 - Tight junctions: CSF components must pass trans-cellular
 - **Extra-choroidal ependymal** cells (ependymocytes)
 - Ciliated ependymal cells (lining ventricular system)
 - Propel CSF through ventricles and spinal cord
 - Gap junctions for exchange between CSF and interstitial fluid



Other non-neuronal cells exist in the brain, such as ependymal cells, which provide the lining of the fluid-filled ventricles. There are two types of ependymal cells, namely *Group 1* choroidal epithelial cells of the choroid plexus and *Group 2* extra-choroidal ependymal cells. Group 1 cells are joined by tight junctions excluding passage of water and solutes by the paracellular route across the choroid plexus, forming the blood-CSF barrier. Group 2 cells (ependymocytes) line the ventricles of the brain and central canal of the spinal cord and are in contact with the cerebrospinal fluid (CSF). They are joined together by gap junctions (not by tight junctions).

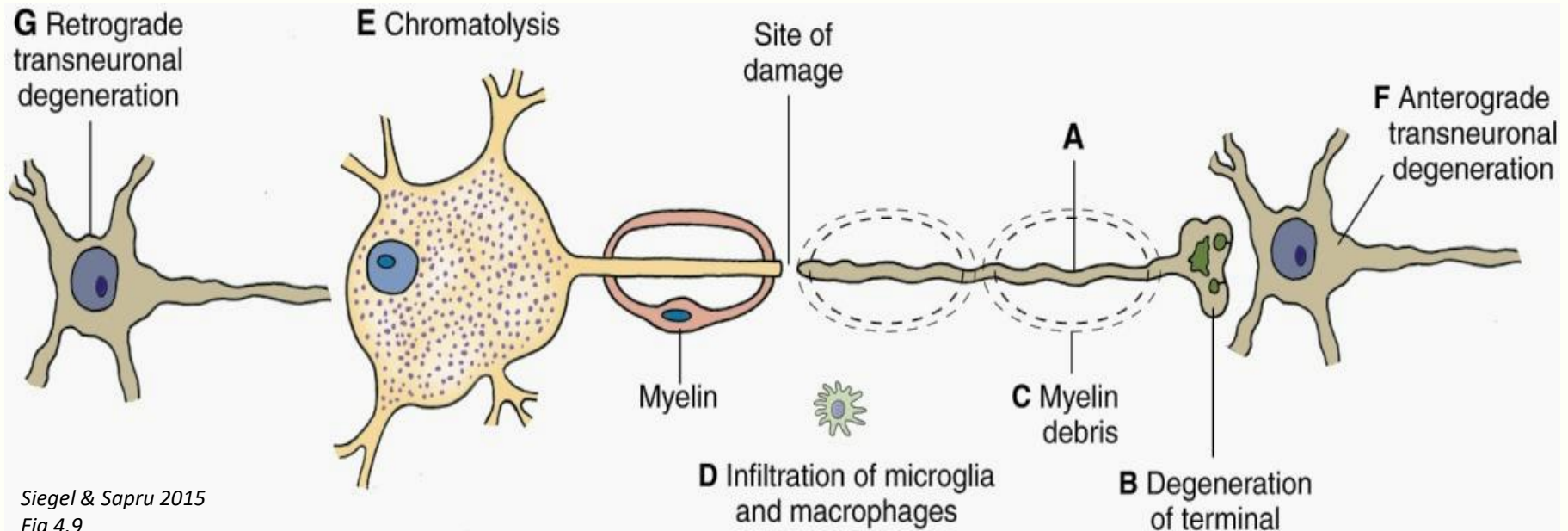
Group 1. These cells are the secretory cells of the Choroid Plexus. Virtually all solutes and water molecules that pass from blood to CSF via the choroid plexus must pass through the cytoplasm of these cells (transcellular route).

Group 2. These cells are ciliated and form the ventricular linings of the brain and spinal cord. They assist with propulsion and circulation of CSF within CNS. Moreover, solutes and water may pass readily around the cells via the paracellular route. This permeable route allows effective exchange between CSF and interstitial fluid.

Lecture Objectives

- SOM.MK.I.BPM2.1.NB.1.NSCI.0010 Identify two major cell types in the nervous system
- SOM.MK.I.BPM2.1.NB.1.NSCI.0011 Identify the dendrites, soma, axon and axon terminals of neurons
- SOM.MK.I.BPM2.1.NB.1.NSCI.0012 Describe functional properties of neuronal organelles
- SOM.MK.I.BPM2.1.NB.1.NSCI.0013 Correlate cytoskeletal units with active transport mechanisms
- SOM.MK.I.BPM2.1.NB.1.NSCI.0014 Correlate neuronal morphotype and intercellular reception/transmission
- SOM.MK.I.BPM2.1.NB.1.NSCI.0015 Correlate conduction velocity with axonal myelination and diameter
- SOM.MK.I.BPM2.1.NB.1.NSCI.0016 Differentiate excitatory and inhibitory neurons
- SOM.MK.I.BPM2.1.NB.1.NSCI.0017 Describe the types of glial cell and their functions in the CNS and PNS
- SOM.MK.I.BPM2.1.NB.1.NSCI.0018 List cellular reactions to axonal transection
- SOM.MK.I.BPM2.1.NB.1.NSCI.0019 Correlate select clinical conditions (tumors of glial origin, gliosis, Guillain-Barre syndrome) with cellular phenomena

Regeneration and Degeneration of Axons

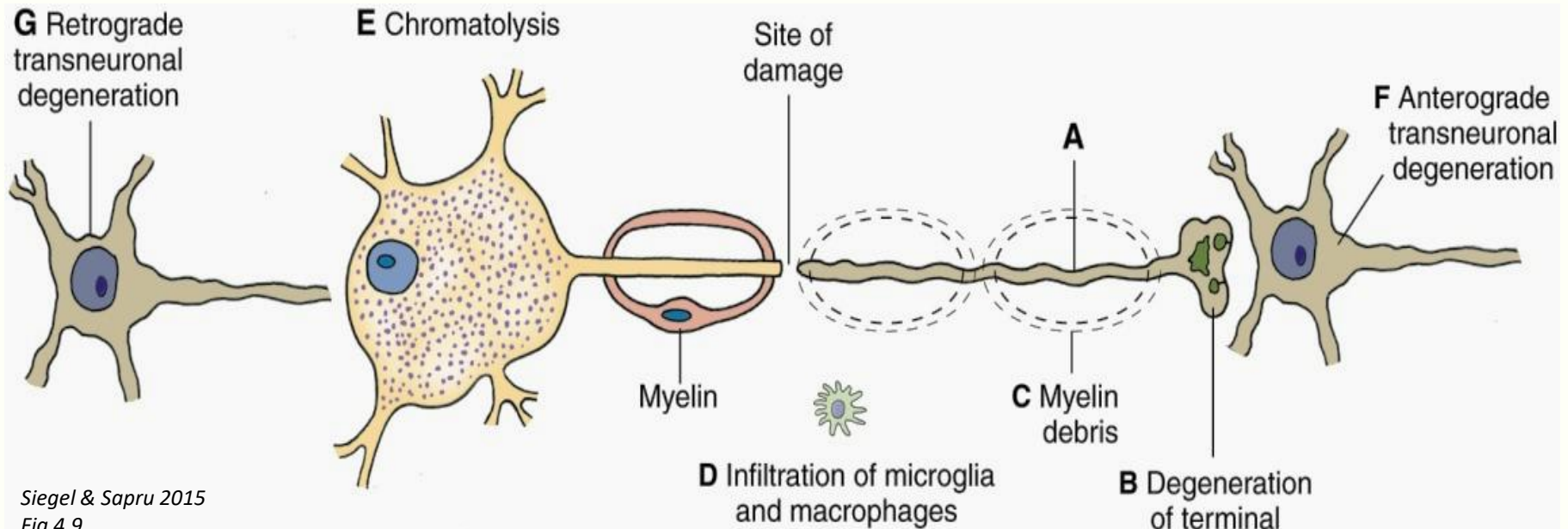


Siegel & Sapru 2015
Fig 4.9

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- | | |
|--|---|
| <ul style="list-style-type: none">A. Wallerian degeneration distal to axonal transectionB. Degeneration of the axonal terminalC. Schwann cells degenerate, leaving a debris fieldD. Microglia infiltrate, phagocytizing | <ul style="list-style-type: none">E. Pumps fail in the traumatized cell, yielding swelling, and dilution of organellesF. Cells no longer receiving synaptic input from the traumatized axonal terminal may degenerateG. Cells no longer providing synaptic input to the traumatized neuron may degenerate |
|--|---|

Regeneration and Degeneration of Axons



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PNS

- Proximal segment of damaged axon may form sprouts that reconnect with peripheral targets (growth ~ 2 mm/day)
- Schwann cells proliferate:
 - Forming a guide tube
 - Releasing nerve growth factors
 - Re-myelinating the regenerated axon

CNS

- Oligodendrocytes and astrocytes inhibit axonal regeneration
 - Do not form guide tubes like Schwann cells
 - Nogo protein blocks axon growth

Regeneration of Axons in the PNS

- Following axonal transection
 - Distal axonal segment degenerates
 - Proximal axonal segment may form sprouts
 - Schwann cells
 - Proliferate, forming a guide tube
 - Release nerve growth factors, encouraging axonal regeneration
 - Re-myelinate the regenerating axon (axons grow at a rate of about 2 mm/day)
 - Target cells may die unless reinnervated

Chromatolysis reflects post-traumatic neuronal swelling and dilution of organelles.

Anterograde degeneration reflects the transsynaptic death of the denervated (postsynaptic) cell.

Retrograde degeneration reflects the death of the presynaptic cell.

Regeneration of transected CNS axons is currently clinically untenable. Central glial cells do not secrete significant amounts of nerve growth factor. Oligodendrocytes do not form guide tubes as do the Schwann cells in the PNS. They secrete 'Nogo' protein which inhibits axonal growth. Moreover, gliotic scars block axonal regrowth.

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Clinical Correlates

- Multiple Sclerosis
 - Autoimmune demyelinating disease of the CNS
 - Blurring, partial loss of vision – CN II
 - Muscle weakness - corticospinal tracts
 - Incoordination – cerebellum
- Guillain-Barré syndrome
 - Inflammatory demyelinating disease of the PNS
 - Affects myelin sheath and axons
 - Symmetrical ascending paralysis
- Tumors of glial origin
 - ~ 50% brain tumors and 25% of spinal tumors